# Laue Diffraction and X-Ray Crystallography

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Laue transmission photographs and powder X-ray diffraction (XRD) were used to test for crystalline order and to identify two unknown solids. A LiF mini-crystal gave about 15 sharp, discrete spots, establishing single-crystal order; the spot positions showed no mirror or rotation symmetry beyond chance, so the beam was not along a low-index axis and the pattern cannot confirm the cubic point group. An undrawn polyethylene monofilament produced an isotropic diffuse ring, which after drawing concentrated into two arcs perpendicular to the filament axis (anisotropy 0.02 before and 0.87 after drawing), consistent with chains aligned along the draw direction. The powder patterns were indexed from ratios of $\sin^2\theta$, with peak positions from profile fits and lattice parameters corrected for a systematic goniometer angle error. Unknown 1 is diamond-cubic silicon, $a = (5.4311 \pm 0.0002)$ Å, identified by the absence of the (200) and (222) reflections that the two-sublattice structure factor forbids. Unknown 2 is body-centred cubic molybdenum, $a = (3.1480 \pm 0.0013)$ Å; its (200) reflection is 24 times stronger than (110), the reverse of the random-powder expectation, which indicates strong preferred orientation.

## 1. INTRODUCTION

Determining the arrangement of atoms in a solid is one of the most direct ways to understand its physical properties. The electrical conductivity of silicon, the ductility of molybdenum, and the mechanical response of a polymer to stretching all follow from how their atoms or molecules are ordered. X-ray diffraction provides this structural information because X-ray wavelengths (roughly 0.5–2.5 Å) are comparable to interatomic spacings, so a crystal acts as a three-dimensional diffraction grating whose pattern reflects the underlying atomic arrangement [1,2].

Two complementary techniques exploit this. In Laue diffraction a polychromatic ("white") X-ray beam illuminates a single crystal. Each set of lattice planes $(hkl)$ selects from the continuous spectrum the wavelength that satisfies the Bragg condition at the angle fixed by the crystal orientation, and the resulting spots on a photographic film carry the symmetry and orientation of the crystal [1]. For partially ordered materials such as drawn polymers, the anisotropy of the scattering instead indicates the degree of molecular alignment.

In powder X-ray diffraction (XRD) a monochromatic beam strikes a polycrystalline sample whose crystallites are randomly oriented. Every set of planes that satisfies the Bragg condition produces a cone of diffracted radiation; scanning the detector angle $2\theta$ converts the cones into discrete peaks whose positions give the interplanar spacings, and whose sequence identifies the crystal structure and lattice parameter [3]. This experiment uses Laue photographs to test for single-crystal order in LiF and for drawing-induced alignment in polyethylene, and powder XRD to identify two unknown cubic solids and measure their lattice parameters.

## 2. THEORY

### 2.1 X-Ray Production

X-rays are produced by two mechanisms, each yielding a different kind of spectrum suited to a different diffraction method [1] (Fig. 1a).

When energetic electrons strike a target anode, they can eject inner-shell electrons from the target atoms. Outer-shell electrons then fill the vacancies, emitting X-rays at discrete energies characteristic of the target element. For a copper anode the dominant emission is the Cu $K_\alpha$ doublet, $K_{\alpha1}$ at $\lambda_1 = 1.540598$ Å and $K_{\alpha2}$ at $\lambda_2 = 1.544426$ Å. This nearly monochromatic radiation is required for powder diffraction, where each peak must correspond to a single well-defined wavelength.

The same electrons also lose energy when deflected in the Coulomb field of target nuclei, emitting bremsstrahlung, a continuous spectrum. Its maximum photon energy equals the electron kinetic energy $eV$ [1], so the spectrum extends to wavelengths above $\lambda_{\min} = hc/(eV)$, which is 0.41 Å for a tube voltage $V = 30$ kV. This white beam is required for Laue diffraction, where the continuous spectrum allows every set of lattice planes to find a wavelength satisfying the Bragg condition.

### 2.2 The Bragg Condition

Waves reflected from successive parallel planes of spacing $d$, incident at glancing angle $\theta$, interfere constructively when their path difference is an integer number of wavelengths [1]:

$$n\lambda = 2d\sin\theta \tag{1}$$

In Laue diffraction the angle is fixed by the crystal orientation and the wavelength $\lambda$ is selected from the white beam; in powder diffraction $\lambda$ is fixed and $\theta$ is scanned.

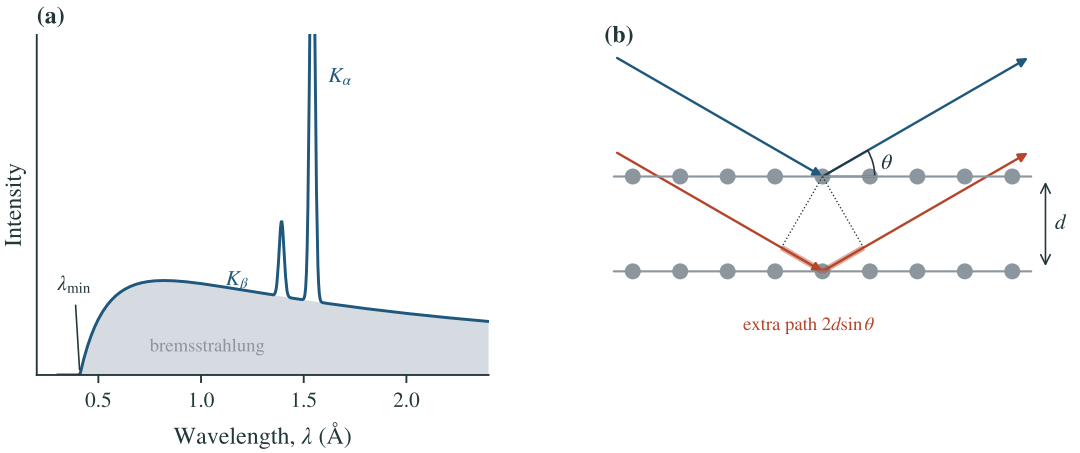



FIG. 1. (a) Schematic spectrum of an X-ray tube with a Cu anode: the bremsstrahlung continuum, which begins at $\lambda_{\min} = hc/(eV)$, and the characteristic $K_\alpha$ and $K_\beta$ lines. (b) Bragg reflection: the ray reflected from the lower plane travels an extra $2d\sin\theta$ (shaded), so the two reflected rays interfere constructively when this equals $n\lambda$.

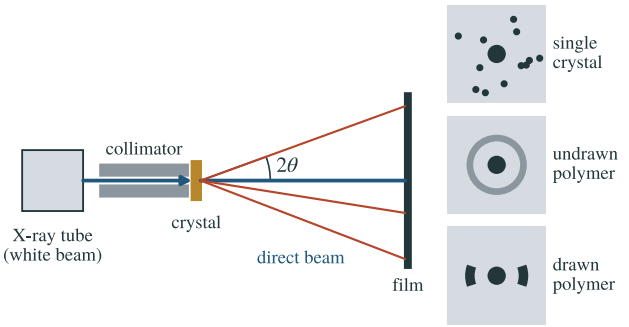



FIG. 2. Laue transmission geometry. The white beam passes through the collimator and the sample; the film records the direct beam at its centre and the diffracted beams around it, each at its own angle $2\theta$. Right: the patterns expected on the film for a single crystal (discrete spots), an unoriented polymer (a diffuse ring) and a drawn polymer with a vertical fibre axis (equatorial arcs).

### 2.3 Laue Diffraction in Transmission

In the arrangement used here the crystal sits on the collimator and the film is placed downstream of it, so the film records the transmitted direct beam at its centre and the beams diffracted forward ($2\theta < 90°$) around it (Fig. 2; Fig. 2 of Ref. [1]). Each spot corresponds to one set of planes, which reflects the wavelength $\lambda_{hkl} = 2d_{hkl}\sin\theta_{hkl}$ from the beam. When the beam lies along a symmetry axis of the crystal, the spot pattern displays that axis's rotational and mirror symmetry; for a general orientation it need not show any.

A polymer gives a different pattern. An unoriented semicrystalline polymer contains small crystallites in all orientations, so its scattering forms diffuse rings around the direct beam. Drawing aligns the chain axes (the crystallographic $c$ axis of polyethylene) along the draw direction, producing a fibre texture. The strongest polyethylene reflections, (110) and (200), come from planes that contain the chain axis, so in a fibre they concentrate into arcs on the equator, the line through the direct beam perpendicular to the fibre axis [4].

### 2.4 Powder Diffraction and Cubic Indexing

For a cubic crystal with lattice parameter $a$, the interplanar spacing is [2,3]

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}, \tag{2}$$

and combining with Eq. (1) (setting $n = 1$ and absorbing higher orders into the Miller indices) gives

$$\sin^2\theta = \frac{\lambda^2}{4a^2}\left(h^2 + k^2 + l^2\right). \tag{3}$$

The ratios of $\sin^2\theta$ for successive peaks therefore reproduce the ratios of the allowed integers $N = h^2 + k^2 + l^2$, which identify the lattice type.

### 2.5 Structure Factor and Selection Rules

The structure factor determines the scattering amplitude from one unit cell [2,3]. Fig. 3 shows the atom positions used below:

$$F(hkl) = \sum_{j=1}^{M} f_j \exp\left[2\pi i\left(hx_j + ky_j + lz_j\right)\right], \tag{4}$$

where $f_j$ is the atomic scattering factor of the $j$-th of $M$ atoms and $(x_j, y_j, z_j)$ are its fractional coordinates. Reflections with $F = 0$ are systematically absent. A simple cubic (SC) lattice

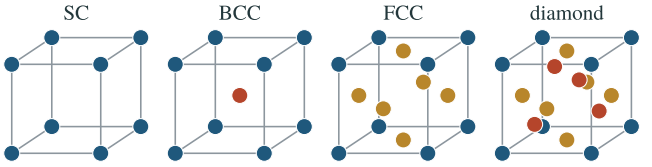



FIG. 3. Conventional cubic unit cells. BCC adds an atom at the cell centre (red) and FCC an atom at each face centre (gold); diamond cubic adds to the FCC cell a second FCC sublattice shifted by (1/4, 1/4, 1/4) (red), four atoms of which lie inside the cell.

has no absences. For a body-centred cubic (BCC) lattice, with atoms at $(0,0,0)$ and $(1/2,1/2,1/2)$,

$$F = f\left[1 + e^{i\pi(h+k+l)}\right] = \begin{cases} 2f & \text{if } h+k+l \text{ is even} \\ 0 & \text{if } h+k+l \text{ is odd} \end{cases} \tag{5}$$

For a face-centred cubic (FCC) lattice, with atoms at $(0,0,0)$, $(1/2,1/2,0)$, $(1/2,0,1/2)$ and $(0,1/2,1/2)$,

$$\begin{aligned} S_{\mathrm{FCC}} &= f\left[1 + e^{i\pi(h+k)} + e^{i\pi(h+l)} + e^{i\pi(k+l)}\right] \\ &= \begin{cases} 4f & \text{if } h,\ k,\ l \text{ all even or all odd} \\ 0 & \text{otherwise} \end{cases} \end{aligned} \tag{6}$$

The allowed values of $N$ are therefore 1, 2, 3, 4, 5, 6, 8, … (SC), 2, 4, 6, 8, 10, 12, … (BCC) and 3, 4, 8, 11, 12, 16, 19, … (FCC).

### 2.6 Diamond-Cubic Structure Factor

The diamond-cubic structure (silicon, germanium, diamond) has the 8-atom basis specified in the manual [3]: an FCC sublattice at $(0,0,0)$, $(1/2,1/2,0)$, $(1/2,0,1/2)$, $(0,1/2,1/2)$ and a second FCC sublattice at $(1/4,1/4,1/4)$, $(3/4,3/4,1/4)$, $(3/4,1/4,3/4)$, $(1/4,3/4,3/4)$. Because the second sublattice is the first shifted by $(1/4,1/4,1/4)$, Eq. (4) factorises:

$$F(hkl) = S_{\mathrm{FCC}} \cdot \left[1 + e^{i\pi(h+k+l)/2}\right]. \tag{7}$$

The second factor, $P = 1 + e^{i\pi(h+k+l)/2}$, adds extinctions beyond the FCC rule (Table I).

Reflections such as (200) ($h+k+l=2$) and (222) ($h+k+l=6$) are therefore forbidden in diamond cubic but allowed in FCC. Their absence distinguishes the two structures.

## 3. PART A: LAUE DIFFRACTION

### 3.1 Methods

The Laue photographs were taken on an enclosed X-ray unit whose scatter shield must be locked before the tube can be energised, following the manual [1]. A LiF mini-crystal was mounted with clear adhesive tape over the aperture of the 1 mm diameter primary-beam collimator (582.002), with its longitudinal axis at 45°, and the collimator was fitted with the auxiliary slide carriage to the lead-glass dome. A 38 mm × 35 mm film in the film cassette (562.013), without a primary-beam shield, was exposed for 40 min at experimental station E.S. 3. The tube settings for this exposure were not recorded.

TABLE I. Structure factor of the diamond-cubic 8-atom basis for FCC-allowed reflections, from Eq. (7). Reflections with $h+k+l \equiv 2 \pmod 4$ are extinguished by destructive interference between the two sublattices.

| $h+k+l \bmod 4$ | $e^{i\pi(h+k+l)/2}$ | $P$ | $\lvert F\rvert^2$ |
|---|---|---|---|
| 0 | 1 | 2 | $64f^2$ |
| 1 | $i$ | $1+i$ | $32f^2$ |
| 2 | $-1$ | 0 | 0 |
| 3 | $-i$ | $1-i$ | $32f^2$ |

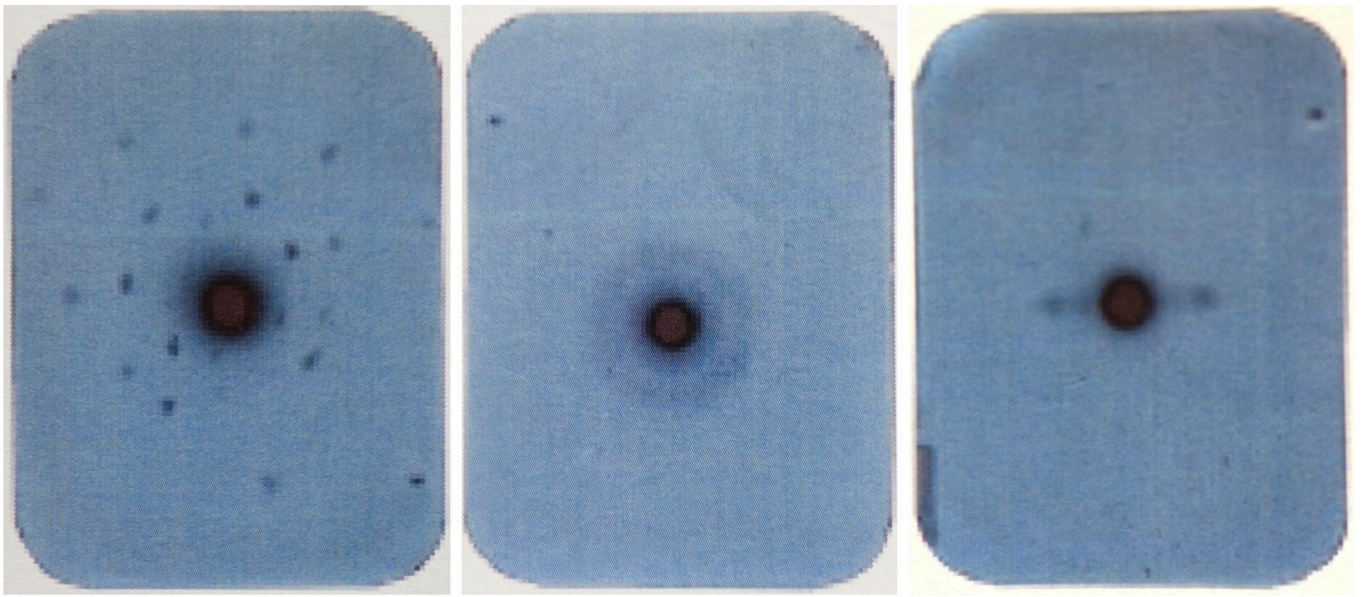
A polyethylene monofilament (585.006) was then mounted vertically over the 1 mm port and exposed for 7 min at E.S. 3 with 30 kV and 75 $\mu$A. Its central section was stretched by hand until it had a clearly drawn form, and a second 7 min exposure was taken with the drawn portion over the port and all other settings unchanged. The developed films were photographed. The spot positions and the azimuthal distribution of film darkening were measured from the photographs; because the photographic response was not calibrated, only positions and relative contrasts are used. The sample-to-film distance was not recorded, so spot positions cannot be converted into scattering angles.

### 3.2 Results and Analysis

The LiF film (Fig. 4a) shows discrete, isolated spots, the signature of a single crystal with long-range periodicity. Spots were counted as compact regions darker than the smoothed background by more than 4–5 robust noise widths and larger than 80 pixels; this gave 15 spots, and 13–18 for thresholds of 3–6 noise widths. To test the symmetry, each spot position was reflected about lines through the direct beam, and rotated by 180°, 120° and 90° about it, and the fraction of spots landing within 20 pixels (1.7% of the film height) of another spot was recorded. The best mirror line matched 40% of the spots and the best two-, three- and four-fold rotations 27%, 27% and 13%. Spots placed at the same radii with random azimuths gave fractions at least as large in 35%, 72%, 18% and 95% of 200 trials. The pattern therefore has no detectable symmetry about the beam.

LiF has the rock-salt structure, with cubic point group $m\bar{3}m$ [2], so a beam along a cube axis would give a four-fold pattern. The absence of any symmetry shows that the beam was not along a low-index zone axis; the tape mount fixed the crystal's in-plane angle but not its tilt. The photograph therefore establishes single-crystal order but cannot confirm the cubic point group.

In polyethylene, drawing changed the scattering from isotropic to strongly anisotropic. The undrawn filament (Fig. 4b) produced a faint diffuse ring with no sharp features. In the drawn filament (Fig. 4c) the ring is replaced by two arcs on opposite sides of the beam, on the horizontal line perpendicular to the vertically mounted filament. To quantify this, the darkening in an annulus of 1.6–3.4 direct-beam radii was averaged within 20° of the horizontal ($H$) and vertical ($V$) directions, after subtracting the median darkening of an outer annulus, and the anisotropy $A = (H-V)/(H+V)$ was formed. The undrawn film gave $A = 0.02$ and the drawn film $A = 0.87$; over five choices of annulus, $A$ ranged from $-0.02$ to 0.12 and from 0.72 to 1.3 respectively ($A > 1$ when $V$ falls below the outer background). Equatorial arcs are where the (110) and (200) reflections of chains aligned along the fibre axis are expected [4], so the change is consistent with drawing-induced chain alignment. The drawn pattern still shows arcs rather than sharp spots, so the crystallites remain small and their orientations spread about the fibre axis.



(a) (b) (c)

FIG. 4. Photographs of the Laue transmission films. The dark central disc in each is the direct beam, which was not blocked. (a) LiF mini-crystal: about 15 discrete spots on a uniform background. (b) Undrawn polyethylene: a faint diffuse ring around the direct beam with no preferred direction. (c) Drawn polyethylene (filament axis vertical): the scattering is concentrated into two arcs on the horizontal line through the beam, perpendicular to the filament axis.

## 4. PART B: POWDER X-RAY DIFFRACTION

### 4.1 Methods

Two unknown polycrystalline samples were scanned on a PANalytical X'Pert$^3$ powder diffractometer [3] in $\theta$–$2\theta$ reflection geometry, with a Cu anode, a Ni $K_\beta$ filter and no monochromator. Both scans covered $2\theta = 20°$–$140°$ in continuous mode at 0.02° per step and 0.40 s per step, with a 1° divergence slit and a 0.10 mm receiving slit. The tube was operated at 40 kV and 30 mA for Unknown 1 and at 45 kV and 40 mA for Unknown 2; the full settings are listed in Table IV.

Each indexed reflection was fitted by weighted least squares over a window extending 0.6° either side of the doublet. The model was a $K_{\alpha1}/K_{\alpha2}$ pair of split pseudo-Voigt profiles (asymmetric widths below and above the maximum, to follow the low-angle tail of the peaks) on a linear background. The $K_{\alpha2}$ position was fixed by Eq. (1) and the wavelength ratio, its intensity was fixed at half that of $K_{\alpha1}$, and the counts were given Poisson weights. Position uncertainties are the fit standard errors, scaled by $\sqrt{\chi^2_\nu}$ where the reduced chi-squared exceeded one; they range from 0.001° to 0.012°. Each $K_{\alpha1}$ position was converted to $a$ with Eq. (3) and $\lambda_1$.

An angle error common to all peaks biases $a$ in a way that depends on $\theta$. Two standard forms were fitted to the peak positions [4]: a constant zero offset, $2\theta_{\mathrm{obs}} = 2\theta + z$, and a displacement $s$ of the sample surface from the goniometer axis, $2\theta_{\mathrm{obs}} = 2\theta - 2s\cos\theta/R$, with goniometer radius $R = 240$ mm. Each fit has $a$ and one error parameter free. A Nelson–Riley extrapolation, a weighted linear fit of $a$ against $\mathrm{NR} = \left(\cos^2\theta/\sin\theta + \cos^2\theta/\theta\right)/2$ extrapolated to NR = 0 [4], was also made. The adopted lattice parameter is the midpoint of the corrected values. Its uncertainty combines the half-range of the corrected values, as the model uncertainty, in quadrature with their scaled fit uncertainty.

### 4.2 Results and Analysis

#### 4.2.1 Unknown 1: Diamond-Cubic Silicon

The Unknown 1 scan (Fig. 5) contains 11 peaks between $2\theta = 28°$ and 137°, and their $\sin^2\theta$ ratios index it as diamond cubic. Multiplied by three, the ratios to the first peak are integers to within 0.06 (Table II): 3, 8, 11, 16, 19, 24, 27, 32, 35, 40, 43. These are FCC-allowed values, but $N = 4$ (200) and $N = 12$ (222) are missing. Both have $h + k + l \equiv 2 \pmod 4$, where $P = 0$ in Eq. (7). For an FCC metal (200) is among the strongest reflections, as in the manual's gold example [3]. Here, a doublet of fixed width fitted at the predicted positions gives heights of $0 \pm 2$ counts for (200) at 32.94° and $-1.5 \pm 1.5$ counts for (222) at 58.84°, less than 0.2% of the (111) height.

The individual lattice parameters decrease steadily with angle, from 5.4346 Å at (111) to 5.4310 Å at (533). The spread, 0.0036 Å, is more than ten times the uncertainty of each value, so it reflects a systematic angle error rather than random scatter. The zero-offset fit gives $z = (-0.020 \pm 0.002)°$ and $a = (5.4309 \pm 0.0001)$ Å; the displacement fit gives $|s| = (0.039 \pm 0.004)$ mm and $a = (5.4312 \pm 0.0001)$ Å; the Nelson–Riley intercept is $(5.4312 \pm 0.0001)$ Å. All three have $\chi^2_\nu \approx 2$. The adopted value is $a = (5.4311 \pm 0.0002)$ Å, which agrees with the silicon lattice parameter, 5.4310 Å at 22.5 °C [5]. The uncorrected mean, $(5.4322 \pm 0.0010)$ Å (standard deviation), is 0.02% higher. Unknown 1 is therefore silicon.

The $K_\alpha$ doublet confirms the wavelength assignment used in the fits. Differentiating Eq. (1) at fixed $d$ gives $\Delta(2\theta) = 2(\Delta\lambda/\lambda)\tan\theta$, with $\Delta\lambda = \lambda_2 - \lambda_1 = 0.003828$ Å. For the five pairs in which both maxima are separately resolved, (331) to

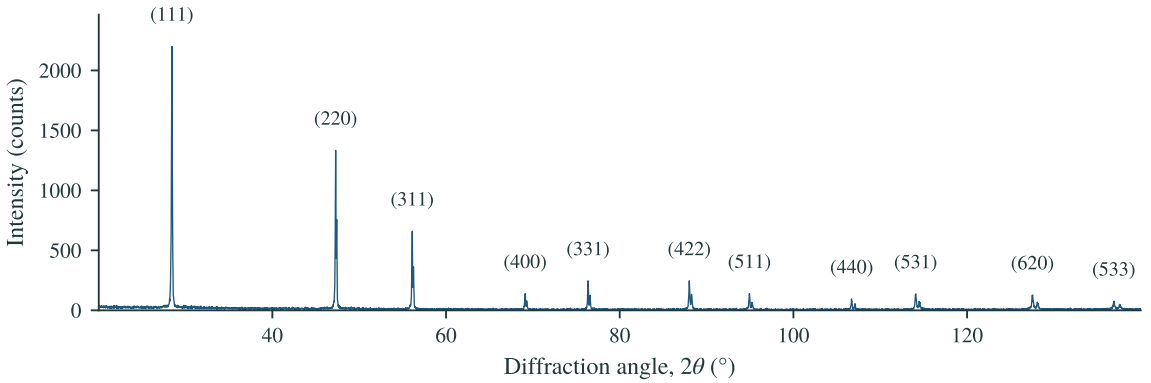



FIG. 5. Powder diffraction scan of Unknown 1 (Cu $K_\alpha$, 0.02° steps), with each peak labelled by its Miller indices. Every reflection has $h, k, l$ all odd or all even with $h + k + l \not\equiv 2 \pmod 4$, the diamond-cubic selection rule; (200) at 32.9° and (222) at 58.8° are absent. Above $2\theta \approx 70°$ each peak is split into its $K_{\alpha1}$ and weaker $K_{\alpha2}$ components.

TABLE II. Indexed reflections of Unknown 1. $2\theta$ is the fitted $K_{\alpha1}$ position (uncertainty 0.001°–0.012°, i.e. 0.0001–0.0003 Å in $a$); "ratio" is $\sin^2\theta/\sin^2\theta_{111}$; height is the fitted $K_{\alpha1}$ peak height above background in counts per step. The decrease of $a$ with angle is the signature of a systematic angle error.

| $(hkl)$ | $2\theta$ (°) | $3 \times$ **ratio** | **Height** | $d$ (Å) | $a$ (Å) |
|---|---|---|---|---|---|
| (1 1 1) | 28.423 | 3.00 | 1897 | 3.1376 | 5.4346 |
| (2 2 0) | 47.283 | 8.00 | 1353 | 1.9209 | 5.4331 |
| (3 1 1) | 56.104 | 11.01 | 649 | 1.6380 | 5.4326 |
| (4 0 0) | 69.105 | 16.01 | 137 | 1.3582 | 5.4327 |
| (3 3 1) | 76.357 | 19.02 | 232 | 1.2462 | 5.4321 |
| (4 2 2) | 88.002 | 24.02 | 241 | 1.1089 | 5.4323 |
| (5 1 1) | 94.937 | 27.03 | 126 | 1.0453 | 5.4316 |
| (4 4 0) | 106.706 | 32.04 | 74 | 0.9601 | 5.4310 |
| (5 3 1) | 114.072 | 35.04 | 125 | 0.9181 | 5.4316 |
| (6 2 0) | 127.523 | 40.05 | 112 | 0.8588 | 5.4314 |
| (5 3 3) | 136.889 | 43.06 | 56 | 0.8282 | 5.4310 |

(620), the measured separations (0.22°, 0.26°, 0.32°, 0.44°, 0.58°) agree with the calculated ones (0.224°, 0.275°, 0.310°, 0.439°, 0.578°) to within one 0.02° step. The $K_{\alpha2}$ to $K_{\alpha1}$ height ratios are 0.46–0.54, consistent with the ratio of 0.50.

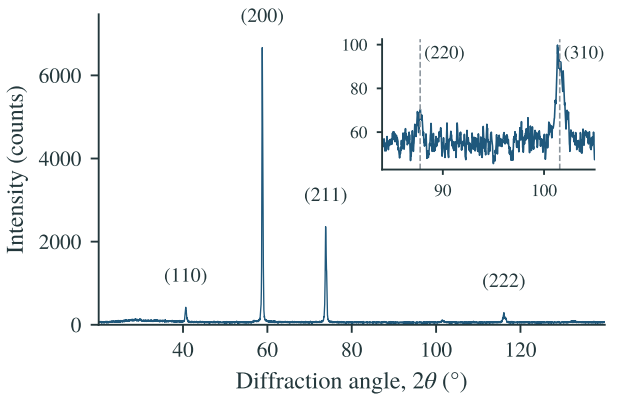



FIG. 6. Powder diffraction scan of Unknown 2, labelled as in Fig. 5. Only reflections with $h + k + l$ even appear, as required for BCC. Inset: the weak (220) and (310) reflections (5-point running mean); dashed lines mark their positions predicted from the fitted lattice parameter. The broad rise between 24° and 34° is diffuse background, not Bragg peaks.

### 4.2.2 Unknown 2: Body-Centred Cubic Molybdenum

The four strong peaks of Unknown 2 (Fig. 6) have $\sin^2\theta$ ratios of $1 : 2.00 : 2.99 : 5.97$, which give $N = 2, 4, 6, 12$ and index them as (110), (200), (211) and (222) of a BCC lattice (Table III). The remaining BCC reflections in the scan range, (220), (310) and (321), were tested at their predicted positions, $2\theta = 87.76°$, 101.55° and 132.72°. Fitting a doublet of fixed position and width there gives peak heights of $14 \pm 3$ and $32 \pm 4$ counts above backgrounds of 55 and 66 counts (inset of Fig. 6). With the width free, the heights are $10 \pm 8$ and $34 \pm 8$ counts. (310) is therefore detected and (220) is marginal. (321) is not detected ($3 \pm 4$ counts). Peak positions alone cannot distinguish BCC from SC with lattice parameter $a/\sqrt{2} = 2.226$ Å: the SC sequence $N = 1, 2, 3, 4, 5, 6$ falls at the same angles as BCC $N = 2, 4, \ldots, 12$. The reflection that would separate them, BCC (321) with no SC counterpart ($N = 7$ is not a sum of three squares), is not detected. The BCC assignment therefore rests on the lattice parameter, which matches BCC molybdenum, whereas no SC element with $a \approx 2.23$ Å is listed in Ref. [2].

The lattice parameters increase with angle, from 3.1380 Å at (110) to 3.1454 Å at (222). The error fits give $z = (+0.20 \pm 0.02)°$, $a = (3.1490 \pm 0.0010)$ Å and $|s| = (0.40 \pm 0.04)$ mm, $a = (3.1471 \pm 0.0008)$ Å, and the Nelson–Riley intercept is $(3.1470 \pm 0.0008)$ Å, with slope $(-0.0052 \pm 0.0006)$ Å. All three have $\chi^2_\nu \approx 20$, so the four broad, textured peaks are not fully described by either error model, and the quoted uncertainties are scaled accordingly. The two models cannot be distinguished, and they differ by 0.0019 Å. The adopted value is $a = (3.1480 \pm 0.0013)$ Å, consistent with BCC molybdenum (3.15 Å [2]). The uncorrected mean, $(3.141 \pm 0.003)$ Å, is 0.2% lower. Unknown 2 is therefore molybdenum.

TABLE III. Indexed reflections of Unknown 2, with quantities as in Table II ("ratio" relative to (110)). Position uncertainties are 0.001°–0.011° (0.00005–0.0007 Å in $a$). NR is the Nelson–Riley function.

| $(hkl)$ | $2\theta$ (°) | $2 \times$ **ratio** | **Height** | **NR** | $a$ (Å) |
|---|---|---|---|---|---|
| (1 1 0) | 40.626 | 2.00 | 248 | 2.51 | 3.1380 |
| (2 0 0) | 58.786 | 4.00 | 5987 | 1.51 | 3.1390 |
| (2 1 1) | 73.824 | 5.99 | 2072 | 1.03 | 3.1417 |
| (2 2 2) | 116.063 | 11.94 | 185 | 0.30 | 3.1454 |

### 4.3 Discussion

The molybdenum sample shows strong preferred orientation: its intensities contradict the random-powder model. All BCC reflections have $|F|^2 = 4f^2$, so for random orientation the intensity follows the multiplicity $p$ times the Lorentz-polarisation factor $L_p = (1 + \cos^2 2\theta)/(\sin^2\theta \cos\theta)$ [4], further reduced at high angle by the falling $f$. The product $pL_p$ is 167 for (110), 90 for (211), 36 for (200) and 25 for (222), so (110) should be the strongest peak. Instead (200) is 24 times stronger than (110), and (211) is 8 times stronger. In $\theta$–$2\theta$ reflection geometry only planes parallel to the sample surface diffract, so the data indicate that the crystallites lie with {100} planes preferentially parallel to the surface. The weak (220) peak, parallel to the weak (110), fits the same picture. Texture redistributes intensity but does not shift peaks, so it does not affect the identification. It does reduce the number of usable reflections, which limits the lattice-parameter fit.

The peak widths bound the crystallite size only weakly. The Si peaks have fitted FWHM 0.07°–0.11° below $2\theta = 100°$, rising to 0.23° at (533). Attributing the entire width to finite crystallite size $L$ in the Scherrer relation, $\beta = K\lambda/(L\cos\theta)$ with $K = 0.9$ [4], gives $L \gtrsim 90$ nm. This is a lower bound, because the instrumental width was not measured separately. The Mo peaks are two to three times wider than Si peaks at similar angles, 0.17° at (200) against 0.08° at Si (311), and 0.41° at (222) against 0.16° at Si (531), although the divergence and receiving slits were the same. The excess is broadening by the sample. It does not vary monotonically with angle (0.30°, 0.17°, 0.22°, 0.41° for (110) to (222)), so it cannot be separated into size and strain contributions from four textured peaks; the Scherrer bound for Mo is $L \gtrsim 30$ nm.

The Unknown 2 scan has a broad rise to a median of 115 counts near $2\theta = 29°$, against 57 counts above 80°. The local maxima of 110–140 counts on it are counting fluctuations that do not index to any phase. A broad maximum of this kind is typical of amorphous scattering, but its source (sample holder or mounting material) was not identified. The Si scan shows no comparable rise.

Systematic angle errors limit both lattice parameters. The lattice parameters of both samples trend with angle, in opposite directions. The differences between the uncorrected means and the adopted values, 0.0011 Å (Si) and 0.0070 Å (Mo), are about eight times the statistical uncertainties of the corrected values, so the accuracy of $a$ is set by the angle-error model, not by counting statistics. The Si corrections agree to 0.0003 Å; for Mo the zero-offset and displacement models differ by 0.0019 Å, and this difference dominates the Mo uncertainty. A sample-height error of 0.4 mm is plausible for a sample that does not sit flush with the holder, but with four peaks it cannot be distinguished from a zero offset.

## 5. CONCLUSION

The LiF Laue photograph showed about 15 sharp, discrete spots, establishing single-crystal order, but no mirror or rotation symmetry beyond chance, so the beam was not along a low-index zone axis and the cubic point group could not be confirmed. Drawing a polyethylene monofilament changed its scattering from an isotropic ring ($A = 0.02$) to two arcs perpendicular to the filament axis ($A = 0.87$, at least 0.72 for any annulus tested), consistent with chains aligned along the draw direction. Powder XRD identified Unknown 1 as diamond-cubic silicon, $a = (5.4311 \pm 0.0002)$ Å, through the absence of the (200) and (222) reflections forbidden by the two-sublattice structure factor. Unknown 2 is BCC molybdenum, $a = (3.1480 \pm 0.0013)$ Å, with strong {100} preferred orientation.

The limitations of the analysis suggest specific improvements. The Mo uncertainty is dominated by the unresolved choice between zero-offset and sample-displacement errors; mixing a standard powder of known lattice parameter (e.g. silicon) into the sample would measure the angle error directly. The texture suppressed (110), (220) and (310) and left four strong peaks; grinding the sample, or spinning it during the scan, would restore the missing reflections and test the BCC indexing further. Measuring a line-width standard on the same slits would separate instrumental from sample broadening and turn the Scherrer bounds into estimates. For the Laue patterns, orienting the LiF crystal so that the beam lies along a cube axis would test for the expected four-fold symmetry, and recording the sample-to-film distance would allow spot positions to be converted into scattering angles.

## APPENDIX: INSTRUMENT SETTINGS

TABLE IV. Powder XRD settings for the two unknowns.

| Parameter | Unknown 1 | Unknown 2 |
|---|---|---|
| Diffractometer | PANalytical X'Pert³ | PANalytical X'Pert³ |
| Goniometer; radius | PW3050/60; 240 mm | PW3050/60; 240 mm |
| Sample stage | Flat, PW3071/xx | Flat, PW3071/xx |
| Anode; $K_\beta$ filter | Cu; Ni, 0.020 mm | Cu; Ni, 0.020 mm |
| $\lambda_1$, $\lambda_2$ | 1.540598 Å, 1.544426 Å | 1.540598 Å, 1.544426 Å |
| Generator | 40 kV, 30 mA | 45 kV, 40 mA |
| Scan ($2\theta$) | 20°–140°, continuous | 20°–140°, continuous |
| Step; time per step | 0.020°; 0.40 s | 0.020°; 0.40 s |
| Divergence slit | Fixed 1° (1.52 mm) | Fixed 1° (1.52 mm) |
| Anti-scatter slit | Fixed 1° (1.52 mm) | Fixed 2° (3.03 mm) |
| Incident mask | 10 mm | 10 mm |
| Receiving slit | 1/16° (0.10 mm) | 1/16° (0.10 mm) |
| Soller slits | 0.04 rad (inc. + diff.) | 0.04 rad (inc. + diff.) |
| Detector | Proportional Xe (point) | Proportional Xe (point) |

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